

HCP Profiling App • Territory Reconfiguration Portal

Cloud Production Architecture Blueprint: Universal GCP Cloud Run Deployment & Global Enterprise Accessibility

Target Infrastructure: Google Cloud Platform (GCP) • Cloud Run (Fully Managed Serverless)

Universal Domain: `https://territory.pmi-marketing.com` (or `*.a.run.app`)

Audience: SFE Leads • System Administrators • Executive Commercial Directors

ERPNext Integration: Direct REST API to `https://dev.pmi-marketing.com`

1. Strategic Shift: Local Mobile Host vs. Universal GCP Cloud Run

Limitations of Current Local Host Model (Port 8765)

NETWORK RESTRICTED

- Local Network Dependency:** Requires laptop and phone to be tethered to the same Wi-Fi SSID or mobile hotspot.
- IP Address Volatility:** Device local IP (`192.168.x.x`) shifts every time the mobile device joins a new network.
- Device State Risk:** If phone enters sleep mode, battery saver, or takes a call, the embedded HTTP server process is terminated.
- Zero Remote Collaboration:** SFE leads in different offices cannot collaborate on the same simulation simultaneously.

Advantages of GCP Cloud Run Universal Hosting

UNIVERSAL ACCESS

- Any Device, Anywhere:** Accessible from any Mac, Windows laptop, iPad, or mobile browser across office LAN, home Wi-Fi, and 4G/5G.
- Permanent HTTPS Domain:** Single, memorable URL (`https://territory.pmi-marketing.com`) with free automated Google SSL.
- Serverless Zero-Cost Idle:** Scales down to zero containers when not in use. Monthly infrastructure cost is virtually **\$0.00** under GCP Free Tier.
- Enterprise SLA & Reliability:** 99.95% cloud availability backed by Google Cloud infrastructure with sub-second container cold starts.

2. System Topology & Network Data Flow

Component	Hosting Location	Protocol / Port	Operational Role & Function
Client Layer	Any Laptop, iPad, Desktop, or Phone	HTTPS / 443	SFE Leads and Admins review data, edit codes, stage clusters, and trigger cutover simulation.
Edge & DNS Layer	Cloudflare / Cloud DNS	CNAME Routing	Routes <code>territory.pmi-marketing.com</code> to Google Cloud Run with DDoS and SSL protection.
App Service Layer	GCP Cloud Run (<code>asia-southeast1</code>)	Serverless Container (8080)	Serves clean pure-text portal UI, executes auto-matcher logic, and proxies authenticated API transactions.
ERPNext Database Layer	<code>dev.pmi-marketing.com</code>	REST API / MariaDB	Stores Master Doctypes: <code>Territory</code> , <code>Sales Person</code> , <code>HCP Account</code> , and executes <code>rename_doc</code> and tree rebuilds.

3. Step-by-Step Technical Implementation Plan (Phases 1 - 3)

PHASE 1
CONTAINER

Package Web Portal into an Ultra-Lightweight Container (Dockerfile)

DOCKER ALPINE

- Create an Alpine-based NGINX container image containing `territory_reconfiguration_portal.html`.
- Configure NGINX to proxy live API calls (`/api/*`) directly to `https://dev.pmi-marketing.com`, solving browser CORS restrictions.

Dockerfile
FROM nginx:alpine
COPY assets/web/territory_reconfiguration_portal.html /usr/share/nginx/html/index.html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 8080
CMD ["nginx", "-g", "daemon off;"]

PHASE 2
ARTIFACT

GCP Project Setup & Artifact Registry Image Repository

GCP BUILD

- Enable Cloud Run (`run.googleapis.com`) and Artifact Registry (`artifactregistry.googleapis.com`) in GCP project.
- Build and tag container image in Singapore region (`asia-southeast1`) for lowest latency to Philippine commercial users.

Terminal Setup Commands
gcloud services enable run.googleapis.com artifactregistry.googleapis.com
gcloud artifacts repositories create territory-portal --repository-format=docker --location=asia-southeast1
gcloud builds submit --tag asia-southeast1-docker.pkg.dev/YOUR_PROJECT/territory-portal/web:v1

PHASE 3
DEPLOY

Deploy to Google Cloud Run (Serverless Production Service)

- Deploy with 256MB RAM and 1 vCPU; configure autoscaling from 0 to 5 instances to guarantee zero idle billing.
- Service automatically receives an enterprise-grade Google HTTPS URL with global edge caching.

Cloud Run Deploy Command
gcloud run deploy territory-portal --image=asia-southeast1-docker.pkg.dev/YOUR_PROJECT/territory-portal/web:v1 --platform=managed --region=asia-south

Custom Domain, Enterprise Security, Mobile Integration & Governance

Domain Routing, Mobile Dual-Mode Fallback, Financial Cost Analysis & Rollout Verification Checklist

Target Domain: `territory.pmi-marketing.com` **SSL Certificate:** Google Managed Auto-Renewing SSL (Zero Maintenance)

Estimated Monthly Cost: \$0.00 / month (Fully absorbed by GCP Free Tier) **Mobile Dual-Mode:** Cloud-First with Local Embedded HTTP Server Fallback

4. Steps for Custom Domain Mapping & Mobile App Integration (Phases 4 - 5)

PHASE 4

DOMAIN

Custom Domain Mapping & Automated Managed SSL

DNS CNAME

- In GCP Cloud Run Console → **Custom Domains** → **Add Mapping** → Select `territory-portal`.
- Enter domain name: `territory.pmi-marketing.com`.
- Add the CNAME record in company DNS provider (Cloudflare, GoDaddy, or Route53) pointing to `ghs.googlehosted.com`.
- Google Cloud automatically provisions, signs, and auto-renews a 2048-bit RSA TLS/SSL certificate.

PHASE 5

MOBILE

Mobile App Dual-Mode Integration (Cloud-First + Offline Fallback)

FLUTTER HYBRID

- Update `TerritoryPortalLauncherScreen` in the Flutter app to display the clean cloud URL: `https://territory.pmi-marketing.com`.
- Automatic Connectivity Detection:**
 - Online (Office, Home, 4G/5G):** Opens the universal GCP Cloud Run URL directly.
 - Remote Offline (Clinics with no signal):** Automatically starts local `TerritoryWebServer` on port 8765 as offline fallback.

5. Enterprise Security & Role-Based Access Control (RBAC)

ERPNext Session & Token Authentication

SESSION VALIDATION

- Session Gatekeeper:** The web app verifies user cookie/session with `/api/method/frappe.auth.get_logged_user` upon launch.
- Role-Gated Mutation:** Only accounts bearing `System Manager`, `Sales Manager`, or `SFE Lead` can execute renames, reparenting, or cutover commits.
- Read-Only for Other Roles:** Unauthorized users attempting to reach the URL are blocked from retrieving masterlist trees.

Google Identity-Aware Proxy (IAP) Option

ZERO TRUST (OPTIONAL)

- Corporate SSO Protection:** Can be placed behind Google Cloud IAP tied to company Google Workspace (`@pims-marketing.com`).
- Zero Internet Exposure:** Unauthorized external requests are stopped at Google's edge proxy before ever reaching the portal container.
- Zero Password Fatigue:** SFE leads log in seamlessly using their corporate Google account single sign-on.

6. Financial & Infrastructure Cost Analysis (Monthly Estimates)

Resource / Service	GCP Free Tier Monthly Allowance	Estimated Monthly Usage (SFE Portal)	Net Monthly Cost
Cloud Run Requests	2,000,000 requests / month free	~10,000 - 25,000 requests / month	\$0.00 (100% Covered)
Compute (CPU / Memory)	360,000 vCPU-seconds • 180,000 GiB-seconds free	~5,000 vCPU-seconds / month	\$0.00 (100% Covered)
Artifact Registry Storage	0.5 GB free storage	0.08 GB (1 Alpine NGINX Image)	\$0.00 (100% Covered)
Managed SSL Certificates	Unlimited Google Managed Certificates	1 Certificate (<code>territory.pmi-marketing.com</code>)	\$0.00 (Always Free)
TOTAL ESTIMATED MONTHLY CLOUD EXPENSE:			\$0.00 USD / Month

7. Comprehensive Delivery Checklist & Verification Matrix

Milestone / Task	Responsible Lead	Verification Method & Pass Criteria	Target Status
1. Dockerize Web Portal	DevOps / Engineering	Local Docker run on port 8080 renders portal with 0 console errors.	READY
2. Artifact Registry Push	DevOps / Engineering	Image successfully pushed to <code>asia-southeast1-docker.pkg.dev</code> .	READY
3. Deploy Cloud Run Service	Cloud Architect	Service deployed with 0 min-instances; HTTPS URL responds in < 800ms.	PHASE 1
4. Configure CNAME DNS	Network Admin	<code>territory.pmi-marketing.com</code> resolves to Google Edge IP with valid SSL lock.	PHASE 2
5. End-to-End Live Simulation	SFE Lead / Admin	Simulate code rename, reparenting, and cutover directly against ERPNext database.	PHASE 3
6. Mobile App URL Release	Mobile Team	Release mobile build with universal cloud launcher and offline fallback toggle.	PHASE 4

Conclusion & Next Step: Deploying the Territory Reconfiguration Portal on Google Cloud Run elevates the platform to enterprise production readiness. It eliminates all tethering frictions, provides universal executive access, guarantees zero data loss via Frappe's `rename_doc` and nested set engines, and incurs zero additional monthly infrastructure costs.